

ML-Guided Directed Evolution to Engineer Thermostable and Alkaliphilic PETases

Background: Polyethylene terephthalate (PET) contributes to over 12% of solid global pollution¹ as one of the world's most widely used plastics. PET accumulation drives the need for degradation strategies, and PET-degrading enzymes (PETases) have emerged as a promising approach. IsPETase—a hydrolase isolated from *Ideonella sakaiensis*—was found to break down PET², but is limited by its thermostability, alkaliphilicity, and catalytic efficiency (CE) for large-scale plastic waste processing. **Directed evolution** (DE) is a protein engineering approach that mimics the natural evolution process in laboratory settings and has been **successful in enhancing enzymatic function**. The process iteratively mutates and screens a baseline protein, using the best variant from each round as the template for the next until the target function is achieved³. While this technique has achieved PETase thermostability⁴, industrial PET degradation requires enzymes that also remain active in alkaline environments to neutralize PET acidic byproducts. However, identifying these functional variants by navigating the combinatorial sequence space for multi-site mutations requires a search space too vast for traditional wet-lab screening. **Computational intervention through machine learning (ML) can accelerate DE studies**. By training predictive models on sequence-function datasets, ML can evaluate millions of candidate variants in-silico, significantly reducing experimental burden⁵. Specifically, large-scale pre-trained *sequence-based* protein language models (pLMs) can capture deep evolutionary patterns and predict functional properties such as molecular function and catalytic activity for proteins identified within biosynthetic gene clusters (BGCs)⁵. *Structure-based* protein ML modeling is an emerging field that can greatly improve functional prediction, accounting for tertiary protein properties like residue-residue interactions (RRIs)⁶. These models can be integrated in a ML-guided DE workflow to physically synthesize and validate computationally-identified PETase variants that are functional in industrial settings through laboratory experimental DE screens.

Intellectual Merit: My proposed closed-loop approach will combine pLMs with a novel epistasis-modeling allosteric graph neural network (aGNN) to map BGCs and co-occurring protein patterns. This will model RRIs as a graph by identifying potential functional, highly-connected *hotspots*—areas whose mutations substantially impact protein activity—to engineer PETase variants optimized for industrial depolymerization that maintain catalytic activity at or above 65°C, achieve high reaction rates (>4.5g/L/hr) on amorphous and semi-crystalline PET, and tolerate alkaline environments (pH>7). The ML-guided pipeline to create, align, and refine protein function predictions via DE can be extended beyond PET to engineer enzymes for plastics lacking natural degradation pathways, including polyurethanes, polyamides, and polystyrene⁷, providing a scalable framework for broader biocatalyst engineering. Failure would suggest the bilateral computational approach is limited in predicting protein function and ability to map long-range spatial interactions, which may drive further research into alternative representations for enzymatic activity such as prioritizing quaternary protein interactions compared to residue-level modeling. **I hypothesize that integrating spatial representations of RRIs into pretrained pLMs will capture distant epistatic constraints missed by sequence data alone, accelerating the identification of industrial-grade, thermostable, alkaline-tolerant plastic-degrading enzymes.**

Aim 1: Computationally pre-screen and validate functional variants. Using a curated metagenomic database made in collaboration with the Arnold Research Group at the California Institute of Technology, I will identify a phylogenetically diverse set of 50 potential PETase sequences, using IsPETase as a query template. These enzymes will be characterized for hydrolytic activity and thermostability to generate a seed labeled sequence-function dataset. To evaluate vast mutational search spaces, I will employ the ESM-2 and ProGen pLMs to generate vector representations of PETase sequences that capture latent functional relationships⁸. I will use the SeqProFT framework to efficiently fine-tune the pLMs for the seed dataset, and adapt and optimize predictions for the PETase family⁹. Concurrently, I will build a dynamic aGNN from baseline ESMFold (EF) single-protein structures, mapping residue connections to model

epistatic interactions through long-distance allosteric networks. By training on sequence data and RRI for the graph edge weights, this model is expected to outperform traditional architectures by better accounting for non-local mutational effects. I will concatenate the pLMs and aGNN predictions into a joint feature vector with Bayesian multi-objective optimization to comprehensively infer CE, pH optimum, and thermostability. I will score candidates using the EHVI⁸ acquisition function based on their likelihood of outperforming current combinations. Then, I will select the next 50-variant batch for synthesis by ranking a randomly sampled library of 10⁶ combinatorial variants generated through simultaneous mutagenesis at the top 10 model-identified hotspots that disproportionally influence alkaline stability. To orthogonally test these candidates, I will simulate alkaline conditions by adjusting residue protonation states to reflect high pH environments. Using computational Molecular Dynamics simulations¹⁰, I will assess thermodynamic stability by calculating change in protein folding free energy to verify variant efficacy and rigorously validate the screening pipeline. By pre-screening these combinations, the model enriches the rate of successful functional variants, reducing the number of experimental screening rounds by >50% compared to unguided iterative screening³.

Aim 2: Optimize identified PETase variants with laboratory DE. The top-ranked variants from each in-silico screening round will be assembled via solid -phase protein synthesis, expressed in an *E. coli* bacterial host, and assessed using a QPix Microbial Colony Picker assay. Since some variants may aggregate or misfold in the host, I will employ a SUMO soluble expression tag to guide proper folding¹¹. In 96-well plates, I will subject the *E. coli* lysates to elevated heat and alkaline pH to isolate thermostable and alkaliphilic PETase variants. I will evaluate enzyme stability against various plastic types, including amorphous and semi-crystalline PET substrates to further simulate industrial processing environments. The quantitative data generated from these screens will serve as ground-truth labels. I will feed back the sequence-function data to fine-tune the pLMs for efficient learning across four full evolutions. I will attempt cryo-EM structural characterization¹² for the top variants from the terminal cycles, if time permits. By replacing EF geometric representations with experimentally-derived RRI from cryo-EM, the updated aGNN weights would be grounded in actual protein structure. With this approach, I aim for a 5-fold increase in CE at conditions of 70°C and pH 10 compared to wild-type IsPETase at its optimal 30°C and pH 7.

Broader Impacts: This project could dramatically increase the scale of plastic degradation through viable industrial processes. Beyond synthesizing PETases, simulated DE screens capable of identifying top performing enzymatic variants could catalyze other important chemical reactions relevant in addressing major global challenges. To enhance public understanding of computational protein engineering, I will produce educational content for Caltech's Science Exchange video series in the Biotechnology and AI sectors. I will also adapt a small part of the project workflow into *Build a Better Enzyme* protein mutation modeling challenges and workshops for undergraduate student hackathons, such as the Hacktech event hosted on Caltech's campus in the spring. The provided outreach with my project will help train the next generation of scientists in applying ML to urgent environmental and biological issues.

References: 1. Soong YV et al. *Bioeng.* 2022;9(3):38. 2. Barclay A & Acharya KR. *Biomolecules.* 2023;13(9):1407. 3. Wittmann B et al. *Curr. Opin. Struct. Biol.* 2021;69:11-18. 4. Stevensen J et al. *ACS Omega.* 2025;10(7):6385-6395. 5. Weissenow K & Rost B. *Curr. Opin. Struct. Biol.* 2025;91:102997. 6. Durairaj et al. *Comput. Struct. Biotechnol. J.* 2023;21:630-643. 7. Dong X et al. *J. Environ. Manag.* 2026;404:129422. 8. Subramanian AM et al. *bioRxiv.* 2025. 9. Zhang S & Liu JK. *arXiv.* 2024. 10. Xi Y et al. *Bioresour. Bioprocess.* 2022;9(1):24. 11. Kuo D et al. *Methods Mol Biol.* 2014;1177:71-80. 12. Liao J. et al. *Commun Chem.* 2026;9(1):95.